

Histopathology:

Neoplastic proliferation: Abnormal and uncontrolled growth of neoplastic (tumor-forming) odontogenic epithelial cells.

Follicular Type

Discrete follicles: Separate and distinct clusters of cells.

Fibrous connective tissue: Tissue composed of strong, fibrous protein providing structural support.

Columnar cells: Tall, pillar-shaped cells.

Stellate reticulum: Star-shaped cells found in the central area.

Microcyst formation: Development of small cyst-like spaces.

Plexiform Type

Interconnecting strands: Linked or network-like structures.

Cystic degeneration: Breakdown or transformation of tissue into cyst-like structures.

Acanthomatous Type

Squamous metaplasia: Change of cell type to a squamous (flat) epithelium.

Keratin formation: Production of keratin, a protective protein.

Epithelial pearls: Rounded collections of keratinized epithelial cells.

Stroma: The supportive tissue surrounding epithelial cells.

Granular Type

Granular eosinophilic appearance: Coarse, grainy appearance with a pinkish-red stain under the microscope.

Basal Cell Type

Basal cell carcinoma: A type of skin cancer arising from basal cells.

Clinical Features

Paraesthesia: Tingling or numbness in a specific area.

Mobility of regional teeth: Movement of teeth due to weakened bone support.

Radiological Appearance

Multilocular radiolucent area: A region with multiple dark spaces on X-rays.

Honeycomb appearance: Resembling the structure of a beehive.

Scalloped margin: Wavy, irregular edges.

Differential Diagnosis

Residual cyst: A cyst left behind after the removal of a tooth.

Lateral periodontal cyst: A cyst located along the side of a tooth's root.

Giant cell granuloma: A tumor-like growth with giant cells.

Traumatic bone cyst: A hollow area in the bone caused by trauma.

Treatment

Curettage: Scraping out the tumor.

Peripheral ostectomy: Removal of bone around the tumor site.

En bloc resection: Complete removal of the tumor with surrounding structures.

Extraneous: Involving Ascending Ramus

Ascending ramus: The upward projection of the mandible (lower jaw).

Histological Features

Parakeratin: A form of keratin with retained nuclei, indicating incomplete keratinization.

Corrugated: Wrinkled or ridged surface.

Uniform thickness: Consistent depth throughout the epithelial lining.

Palisaded basal cell layer: Basal cells arranged in a picket-fence-like or tombstone pattern.

Polarized cells: Cells aligned in a specific direction.

Orthokeratin: Fully keratinized cells without retained nuclei.

Connective tissue: Tissue that supports, connects, or separates different tissues and organs.

Satellite cysts: Smaller cysts present within or around the main lesion, associated with a higher recurrence rate.

Lumen: The central cavity or hollow space inside a cyst or structure.

Straw-colored fluid: Yellowish fluid resembling straw, often indicative of protein content.

Protein content: The amount of protein within a fluid, measured in grams per 100 milliliters.

Rete ridges: Extensions or downward growths of the epithelium into the connective tissue.

Differential Diagnosis

(No further details provided in your text for explanation.)

Syndrome Associated with OKC: Gorlin and Goltz Syndrome

OKC (Odontogenic Keratocyst): A type of jaw cyst arising from odontogenic epithelial cells.

Basal cell nevus syndrome: A genetic condition involving skin tumors and jaw cysts.

Autosomal dominant: A genetic inheritance pattern where only one copy of a mutated gene is needed to express the trait.

Features

Multiple OKCs: Presence of several odontogenic keratocysts in the jaw.

Basal cell carcinoma: A form of skin cancer originating in basal cells.

Frontal bossing: Prominent or protruding forehead.

Temporal bossing: Prominence of the sides of the head near the temples.

Hypertelorism: Abnormally increased distance between the eyes.

Mandibular prognathism: Forward projection of the lower jaw.

Calcium/phosphate metabolism: The body's process of regulating calcium and phosphate levels, crucial for bone health.

Calcification of falx cerebri: Hardening or deposition of calcium in the falx cerebri (a membrane dividing the brain hemispheres).

Indications for Biopsy

Lesion: An abnormal area of tissue or damage in the body.

Clinically: Based on physical examination and symptoms.

Superficial ulceration: A shallow sore or erosion on the surface of tissue.

Neoplasms: New, uncontrolled growth of cells, potentially forming tumors.

Types of Biopsy

Excisional biopsy: Complete removal of the lesion for examination.

Incisional biopsy: Partial removal of the lesion for sampling.

Fine Needle Aspiration Cytology (FNAC): Using a thin needle to extract tissue or fluid for analysis.

Frozen section biopsy: A rapid diagnostic method using frozen tissue slices.

Punch biopsy: A technique using a circular blade to extract a tissue sample.

Oropharyngeal region: The middle part of the throat behind the mouth.

Biopsy Technique

Iodine: A chemical used to stain and disinfect the biopsy area.

Antiseptic: A substance that prevents infection by inhibiting microorganism growth.

Adrenaline: A hormone and medication used to reduce bleeding and prolong anesthesia.

Scalpel: A small, sharp surgical instrument used for cutting tissue.

Forceps: A tweezer-like instrument used to hold or grasp tissues.

Fixative: A chemical (e.g., formalin or alcohol) used to preserve tissue samples.

Glazed paper: A smooth, non-absorbent paper used to support thin specimens.

Histopathologist: A medical specialist who examines tissue under a microscope to diagnose diseases.

Labelled: Properly marked with identifying information.

A Benign Multifocal Tumor of Osteoclasts

Benign: Non-cancerous.

Multifocal: Occurring in multiple locations.

Osteoclasts: Bone cells responsible for breaking down bone tissue.

Bone remodeling: The continuous process of bone breakdown and rebuilding.

Aetiology (Cause)

Inflammatory: Related to the body's immune response to injury or infection.

Autoimmune disorders: Conditions where the immune system attacks the body.

Vascular disorders: Issues with blood vessels.

Paramyxovirus: A type of virus affecting the respiratory system or other tissues.

Clinical Features

Vascularity: Rich supply of blood to a tissue.

Platybasia: Flattening or softening of the skull base.

Neurological issues: Problems affecting the brain, spinal cord, or nerves.

Gait difficulties: Problems with walking.

Lion-like facies (Leontiasis ossea): A facial deformity resembling a lion due to bone overgrowth.

Oral Manifestations

Alveolar ridges: Bony ridges in the jaw that hold the teeth.

Cementosis: Excessive deposition of cementum (a calcified layer covering tooth roots).

Dry socket: A painful condition following tooth extraction due to a blood clot dislodging.

Hyperecementosis: Excessive cementum formation around tooth roots.

Ankylosed teeth: Teeth fused to the surrounding bone, preventing movement.

Edentulous: Without natural teeth.

Neuralgias: Sharp, shooting nerve pain.

Laboratory Findings

Serum alkaline phosphatase: An enzyme elevated during bone formation.

Hydroxyproline: A marker indicating collagen breakdown in bones.

NTX/CTX (bone resorption markers): Molecules released during bone breakdown.

Radiological Features

Deossification: Loss of bone mineral density.

Reossification: Abnormal bone rebuilding.

Cotton wool appearance: Radiological pattern resembling fluffy cotton.

Osteoporosis: Reduced bone density and increased fragility.

Osteoblastic activity: Bone formation by cells called osteoblasts.

Root resorption: Breakdown or shortening of tooth roots.

Lamina dura: The thin, dense layer of bone lining tooth sockets.

Treatment: Oral Candidiasis

Polyene Agents: A class of antifungal medications.

Nystatin: An antifungal used to treat yeast infections.

Amphotericin B: A potent antifungal used for severe infections.

Imidazole Agents: Antifungal drugs inhibiting fungal cell membrane synthesis.

Clotrimazole: An antifungal used for treating fungal infections in the skin and mucous membranes.

Ketoconazole: An antifungal used for systemic and skin infections.

Triazoles: Antifungal agents that inhibit fungal growth by blocking ergosterol synthesis.

Fluconazole: A widely used antifungal for systemic infections.

Itraconazole: An antifungal medication for a variety of fungal infections.

Iodoquinol: A drug with antifungal and antibacterial properties.

Systemic Antifungal Treatment: Treatment involving oral or intravenous antifungals to treat widespread infections.

Immunosuppressed: A state in which the immune system is weakened, making the body more susceptible to infections.

Corticosteroids: A class of steroids that reduce inflammation and suppress the immune system.

Angular cheilitis: Inflammation at the corners of the mouth, often due to a fungal infection.

Clinical Features

Pseudomembranous oral candidiasis: A type of oral yeast infection characterized by a white, patchy coating.

Adherent white plaques: White, sticky patches on the surface.

Cottage cheese or curdled milk: Descriptive terms for the appearance of the patches.

Scraping with a tongue blade: Using a flat tool to gently remove the patches.

Erythema: Redness of the skin or mucous membranes due to inflammation.

Atrophic area: A region with reduced tissue thickness or function.

Deeper invasion: Penetration of the infection deeper into tissue layers.

Ulcerative lesions: Open sores or ulcers formed after the infection invades deeper.

Broad-spectrum antibiotics: Medications that target a wide range of bacteria, often disrupting normal microbial balance.

AIDS: Acquired Immunodeficiency Syndrome, a condition that weakens the immune system.

Symptoms

Burning sensation: A painful, stinging feeling.

Foul taste: An unpleasant or bad taste in the mouth.

Erythematous: Reddened due to inflammation.

Atrophic: Loss of tissue thickness or mass, often appearing thin or shriveled.

Histological Features

Histopathologic pattern: The appearance of tissue under a microscope as affected by disease.

Mycelia: The branching, thread-like structures of fungi.

Necrosis: The death of cells or tissue.

Definition

Aetiology: The study of causes or origins of diseases.

Inflammatory cyst: A fluid-filled sac formed due to inflammation.

Epithelium: A tissue type that lines the surfaces and cavities of the body.

Apical periodontal cyst: A cyst at the tip (apex) of the root of a non-vital tooth.

Periapical cyst: A cyst located around the root tip of a tooth.

Non-vital tooth: A tooth that is no longer living, typically due to infection or trauma.

Pulp canal: The central space inside a tooth that contains nerves and blood vessels.

Trauma: Physical injury or damage to the tooth.

Periapical granuloma: A mass of inflamed tissue at the tip of a tooth's root.

Rests of Malassez: Groups of epithelial cells that persist in the tooth's periodontal ligament.

Liquefaction necrosis: Cell death that leads to the transformation of tissue into a liquid form due to lack of nutrients.

Cystic cavity: A fluid-filled space or pocket formed in tissue.

Histopathological Features

Stratified squamous epithelium: A type of epithelial tissue made up of multiple layers of flat cells.

Hyaline bodies: Glass-like structures often seen in tissues under a microscope, associated with cell damage.

Rushton bodies: Special structures found in certain cysts, often containing keratin.

Cholesterol clefts: Small holes or spaces left after cholesterol crystals dissolve in tissue.

Multinucleated giant cells: Large cells with more than one nucleus, often present in chronic inflammation.

Collagen fibers: Strong, fibrous proteins that provide structure to tissues.

Fibroblasts: Cells that produce collagen and other fibers in connective tissue.

Lumen: The hollow space within a cyst or tube-like structure.

Low protein fluid: Fluid inside the cyst that contains fewer proteins compared to normal bodily fluids.

Keratin: A tough protein found in skin, hair, and nails, and occasionally in cysts.

Clinical Findings

Asymptomatic: Showing no symptoms or discomfort.

Septic tooth: A tooth infected with bacteria, often leading to an abscess.

Buccal swelling: Swelling on the cheek side of the face, often due to infection or inflammation.

Crackling sensation: A feeling caused by air or fluid moving under the skin, often due to pressure on a swollen area.

Resorbed bone: Bone that has been broken down and absorbed by the body.

Fluctuation: A wave-like movement felt when pressure is applied to a fluid-filled area.

Bluish color under mucous membrane: A sign of blood pooling under the skin, often due to inflammation or infection.

Sinus opening: A small opening through which pus or fluid may drain from an infection.

Leukoplakia and Histologic Changes

Keratinization: The process of forming a tough, protective layer of keratin on the surface of epithelial cells.

Hyperorthokeratinization: An increase in the thickness of the orthokeratin layer, a normal keratinized layer of epithelium.

Hyperparakeratinization: An abnormal increase in keratin in cells, with nuclei still present in the keratinized layer.

Epithelial dysplasia: Abnormal development or growth of epithelial cells, often a precursor to cancer.

Orthokeratin: A form of keratin found in fully keratinized epithelial cells, without nuclei.

Parakeratin: A form of keratin found in epithelial cells that still retain their nuclei.

Changes in the Cellular Layer

Polarity: The arrangement of cells with distinct structural orientations, especially in basal cells.

Nuclear-cytoplasmic ratio: The proportion between the size of the nucleus and the cytoplasm in a cell.

Scanned with OKEN Scanner: Likely referring to a tool used for scanning or imaging tissue samples, though "OKEN Scanner" might be a specific or proprietary term.

Epithelial stratification: The arrangement of different layers of cells in the epithelium.

Cellular pleomorphism: The presence of a variety of different shapes and sizes of cells, often seen in cancerous or abnormal tissues.

Nuclear hyperchromatism: Darkening of the cell nucleus, indicating increased DNA content, commonly associated with malignancy or abnormal growth.

Cellular cohesion: The ability of cells to stick together or remain connected, often disrupted in dysplasia.

Enlarged nucleoli: Larger-than-normal nucleoli, often indicative of increased cellular activity or malignancy.

Mitotic figures: Visible signs of cell division, often observed in tissues with high cellular turnover or dysplasia.

Thickness of the Epithelium

Epithelial atrophy: A decrease in the size or thickness of the epithelial layer, often due to cell loss or degeneration.

Acanthosis: Thickening of the epidermis or epithelial layer, often due to an increase in the number of cells.

Alteration in Connective Tissue

Destruction of collagen fibers: Breakdown of collagen, a structural protein in connective tissue, often seen in chronic inflammation.

Chronic inflammatory cell infiltrate: The presence of immune cells, such as lymphocytes and macrophages, in connective tissue due to ongoing inflammation.

Connective tissue stroma: The supportive tissue framework that provides structure to organs and tissues.

Definition

Precancerous condition: A condition where changes in tissues are likely to lead to cancer, though not yet cancerous.

Fibrosis: The thickening and scarring of connective tissue, often due to injury or inflammation.

Elasticity: The ability of a tissue to return to its original shape after being stretched or compressed.

Synonyms

Atrophia idiopathica: A term referring to a condition characterized by degeneration or wasting of tissue without a known cause.

Idiopathic scleroderma of the mouth: A form of scleroderma (a connective tissue disease) affecting the oral cavity, with an unknown cause.

Idiopathic palatal fibrosis: Fibrosis in the palate (roof of the mouth) with an unknown cause.

Sclerosing stomatitis: A condition involving inflammation and fibrosis in the mouth.

Submucosal fibrosis of the palate: Fibrosis occurring beneath the mucous membrane of the palate.

Aetiology

Areca nut: A seed of the areca palm, commonly chewed in some cultures, known to cause oral health issues.

Nutritive deficiency: A lack of essential nutrients in the diet.

Immunological factors: Factors related to the immune system, which may contribute to disease development.

Genetic factors: Inherited traits or predispositions that contribute to the development of the condition.

Protracted tobacco use: Long-term use of tobacco, which can lead to various health problems, including OSMF.

Pathogenesis

Hypersensitivity reactions: An exaggerated immune response to a substance that is normally harmless.

Malignant transformation: The process by which normal cells become cancerous.

Clinical Features

Trismus: A condition characterized by restricted mouth opening, often due to muscle spasms.

Leathery lips: Lips that feel thick or stiff, often due to fibrosis.

Microcheilia: Difficulty or inability to turn the lips outward (lip eversion).

Referred pain: Pain that is perceived at a location different from its source.

Histopathology

Hyperkeratinized: An increase in the thickness of the keratin layer of the skin or mucous membranes.

Atrophic epithelium: Thinning or wasting away of the epithelial layer.

Rete pegs: Finger-like projections of the epithelium into the underlying tissue.

Epithelial dysplasia: Abnormal development or growth of epithelial cells, which may be a precursor to cancer.

Nuclear pleomorphism: Variability in the size and shape of the cell nuclei, often a sign of malignancy.

Intercellular edema: Swelling between cells, often indicating inflammation.

Congested blood vessels: Blood vessels that are swollen due to an excess of blood, often due to inflammation.

Homogenization and hyalinization of collagen fibers: The process of collagen fibers becoming

uniform and glassy, a sign of fibrosis.

Perivascular fibrosis: The thickening of tissue around blood vessels, often seen in chronic inflammation.

Treatment

Intralesional injections: Injections made directly into the lesion or affected area to treat the condition.

Collagenase: An enzyme used to break down collagen, which can help in treating fibrosis.

Fibrinolysin: An enzyme used to break down fibrin, which helps in resolving fibrosis and inflammation.

Systemic steroids: Steroid medications that are taken orally or by injection to reduce inflammation throughout the body.

Aetiology

Immune-related disorder: A condition where the body's immune system mistakenly attacks its own cells or tissues.

Exacerbations: Worsening or flare-ups of a condition.

Remissions: Periods of improvement or absence of symptoms.

Emotional trauma: Emotional stress or distress caused by significant life events.

Clinical Features

Middle-aged and elderly: People between 45 and 65 years old (middle-aged) and those 65 or older (elderly).

Female predilection: A tendency for a condition to be more common in females.

Buccal mucosa: The inner lining of the cheeks.

Vestibule: The space between the lips/cheeks and teeth.

Gingiva: The gums or soft tissue surrounding the teeth.

Burning sensation: A painful, stinging feeling in the affected area.

Lesion characteristics: The specific appearance and texture of a medical abnormality or wound.

Wickham's striae: A pattern of tiny white elevated dots or lines seen in oral lichen planus, often arranged in a lace-like network.

Patterns

Linear: Lesions arranged in straight lines.

Papular: Lesions consisting of small, raised bumps.

Reticular: Lesions forming a net-like or mesh pattern.

Annular: Lesions shaped like rings or circles.

Vesicular: Lesions containing small blisters or sacs filled with fluid.

Bullous: Lesions with large blisters.

Erosive: Lesions that cause the skin or mucosa to break down or wear away.

Atrophic: Lesions characterized by thinning or loss of tissue.

Hypertrophic: Lesions characterized by excessive tissue growth.

Histopathology

Hyperorthokeratosis: Thickening of the outer layer of skin cells, with no retention of the nucleus.

Parakeratosis: A condition where the cells in the outer layer of skin retain their nuclei, which is typically not seen in normal skin.

Granular cell layer: The layer of cells beneath the keratin layer of the skin or mucosa, important in protecting tissues.

Acanthosis: Thickening of the skin or mucosal layers due to an increase in the number of cell layers.

Spinous layer: The layer of cells just beneath the granular layer in the skin or mucosa, important for providing strength.

Saw-tooth appearance: A pattern of jagged or irregular projections in the tissue under a microscope.

Necrosis: The death or decay of cells or tissue.

Liquefaction degeneration: The process by which cells or tissue turn into a liquid form due to injury or inflammation.

Band-like subepithelial infiltrate: A dense accumulation of immune cells just beneath the epithelial layer.

T cells: A type of white blood cell involved in immune responses.

Histiocytes: Immune cells involved in inflammation and tissue repair.

Civatte bodies: Eosinophilic (staining intensely with certain dyes) cell remnants seen in degenerating basal keratinocytes.

Max-Joseph spaces: Gaps or clefts seen in histopathology, resulting from the degeneration of basal keratinocytes.

Clinical Features of Radicular Cyst

Periapical lesions: Abnormal growths or changes that occur around the apex (tip) of the tooth root.

Acute apical periodontitis: Inflammation of the tissues around the apex of the tooth, often due to infection.

Periapical abscess: A localized collection of pus around the tooth root due to infection.

Periapical granuloma: A mass of inflamed tissue around the root of a tooth, often resulting from chronic infection.

Periapical cyst: A fluid-filled sac that forms around the apex of a non-vital tooth.

Dentigerous cyst: A cyst that surrounds the crown of an unerupted or partially erupted tooth.

Periapical scar: A healing tissue after an injury or infection around the tooth root.

Giant cell granuloma: A non-cancerous lesion that involves the growth of giant cells, typically found in the jaw.

Osteomyelitis: Infection of the bone, often leading to inflammation and damage.

Periapical cemental dysplasia: A condition where abnormal cementum (the tissue covering the tooth root) forms around the tooth apex.

Langerhans cell disease: A disorder involving abnormal proliferation of Langerhans cells (immune cells), often affecting bones or other organs.

Meaning of Hard Words

Periapical: Referring to the area around the apex (tip) of a tooth root.

Apical: Pertaining to the apex, or tip, of the tooth root.

Periodontitis: Inflammation of the tissues surrounding the teeth, often due to infection.

Abscess: A collection of pus caused by infection.

Granuloma: A mass of tissue that forms as a response to chronic inflammation or infection.

Cyst: A closed sac filled with fluid or semi-solid material.

Scar: A mark left on the tissue after healing from an injury or infection.

Osteomyelitis: Infection or inflammation of bone tissue.

Dysplasia: Abnormal growth or development of cells or tissues.

Langerhans cells: A type of immune cell found in the skin and mucous membranes, involved in immune response.